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MILITARY SPECIFICATION

STEERING SYSTEM, ELECTROHYDRAULIC, MARINE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrohydraulic marine steering systems that comprise motor-driven, electrically controlled, variable-delivery hydraulic pumps powering hydraulic actuators on frigate and larger size Navy ships.

1.2 Classification. Electrohydraulic steering systems are of the following types, as specified (see 6.2):

Type I - Links and tillers.
Type II - Rapson slides.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.
MIL-P-2911 - Packing Assembly, Hydraulic Conical and V Types.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-C-15730 - Coolers, Fluid, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-M-17060 - Motor, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-I-17244 - Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 inch dial).
MIL-M-17508 - Mounts, Resilient, Types 6E2000, 6E900, 6E900BB, 7E450, 7E450BB, 6E150 and 6E100.
MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited.
MIL-P-17869 - Pumps and Motors, Power, Oil Hydraulic (Naval Shipboard Use).
MIL-S-18396 - Switch, Meter and Control, Naval Shipboard Use.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-V-24695 - Valve, and Hose Assembly, Vent and Test Hydraulic Service General Specification for. (Metric)
MIL-F-24702 - Filter Elements, Hydraulic, Disposable, General Specification for. (Metric)
MIL-F-24702/1 - Filter Elements, Hydraulic, Disposable. (Metric)
MIL-T-31000 - Technical Data Packages, General Specification for.
MIL-R-39023 - Resistors, Variable, Non-wirewound, Precision General Specification for.

STANDARDS

MILITARY

MIL-STD-22 - Welded Joint Design.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-419 - Cleaning, Protecting, and Testing Piping, Tubing, and Fittings for Hydraulic Power Transmission Equipment.

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MILITARY (Continued)

MIL-STD-471 - Maintainability/Verification/Demonstration/ Evaluation.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Naval Surface Ships.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-882 - System Safety Program Requirements.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Section 300, Electric Power, Alternating Current.
MIL-STD-1689 - Fabrication, Welding, and Inspection of Ships Structure.
MIL-STD-2193 - Hydraulic System Components, Ship. (Metric)
MIL-STD-45662 - Calibration Systems Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

810-1385917 - Thermometer Selection Guide (Methods of Mounting).

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVPERS 18068 - Manual of Enlisted Manpower and Personnel Classifications and Operational Standards.

ROME AIR DEVELOPMENT CENTER (RADC)

NPRD-2 - Nonelectronic Parts Reliability Data.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 62 - Composition Bronze or Ounce Metal Castings, Specification for. (DoD adopted)

D 3951 - Commercial Packaging, Standard Practice for.

E 208 - Standard Method for Conducting Drop-Weight Test to Determine Nil-Ductility Transition Temperature of Ferritic Steels.

F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

HIR 1063 - General Environmental Considerations for Marine Vehicles.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Design.

3.2.1 General. The steering system shall comprise one or two machinery subsystems and a control subsystem.

3.2.1.1 Machinery subsystem. Each machinery subsystem shall move its associated rudder(s) from the maximum working angle specified (see 6.2), port or starboard, to within 5 degrees of the maximum working angle on the other side, given a sudden hard-over to hard-over change in the helm, at the average rate specified (see 6.2), and shall move, stop and hold the rudder(s) at any angle up to and including the maximum working angle, without exceeding 18 MPa (2600 lb/in^2)

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maximum design working pressure (MDWP), when the maximum rudder torque is as specified (see 6.2). Each machinery subsystem shall operate the rudder(s) smoothly without jiggle, chatter, overshoot or limit cycling. Each machinery subsystem shall comprise the following major components:

Rudder actuator assembly
Two hydraulic power units, each comprised of:
- Main pump
- Main pump motor and controller
- Supply and head tanks
- Servo and return filters
- Replenishment accumulator
- Cooler
- Transfer valve and manifold

Storage tank*
Fill and drain pump unit*
Hand steering pump*

*One of each to be provided per ship

3.2.1.2 General requirements. Each machinery subsystem shall move the rudder(s) to the helm angle ordered by the control subsystem. Each power unit shall be capable of continuously operating the rudder(s), as specified, when running in an active mode. When in a standby mode, a power unit shall be capable of being stopped or running idle with the main pump bypassed. A standby power unit running idle shall transfer to the active mode automatically when the active power unit is normally shut down, and when the faults specified herein shut down the active unit automatically. A stopped standby pump shall start automatically when a failure in the active power unit causes the active power unit to shut down, except as specified herein. There shall be no failure mode in the control subsystem, excluding loss of all ship electrical supply power, that results in simultaneous shutdown of both power units. An electrically powered fill and drain pump shall be provided for filling and flushing the hydraulic power units and for emergency steering. A hand steering pump shall also be provided.

3.2.1.3 Control subsystem. The machinery subsystem shall be designed to interface with the steering control subsystem to meet the requirements specified herein. The local control units and motor controllers shall provide the interface between the control and machinery subsystems (see 6.6).

3.2.1.3.1 Rudder positioning accuracy. Graduations on the mechanical rudder angle indicators in the steering gear room as compared to graduations on the mechanical helm angle indicator of the control unit initiating the order shall have a comparative error of not more than plus or minus 1/4 degree from 0 to 5 degrees of rudder angle and plus or minus 1/2 degree from 5 to 35 degrees of rudder angle.

3.2.1.4 Environmental conditions. The steering system shall start and operate without damage or erratic operation when the ambient temperature within the ship and the initial steering system equipment temperatures are between 0 and 49 degrees Celsius (°C) (32 and 120 degrees Fahrenheit (°F)). The steering system

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shall meet all requirements specified herein when the ambient temperature within the ship and the initial steering system equipment temperatures are between 16 and 49°C (60 and 120°F) except where specified otherwise herein, and when the ship is permanently inclined as follows:

(a) When the ship is permanently trimmed down by the bow or stern as much as 5 degrees or is pitching plus and minus 10 degrees from the normal horizontal plane.
(b) When the ship is permanently listed up to 15 degrees or rolling up to 45 degrees to either side of vertical.

3.2.1.4.1 Nonoperating temperatures. The machinery subsystem equipment shall not be damaged and shall meet all requirements specified herein when restored to the operating temperature range, as specified in 3.2.1.4, after long-term exposure to nonoperating ambient temperature extremes of minus 20°C and plus 65°C (minus 4°F and plus 149°F).

3.2.1.4.2 Electric power requirements. The steering system shall be designed for operating with type I electrical power in accordance with DOD-STD-1399, section 300.

3.2.1.5 Casualty recovery. The steering system shall be so designed and constructed that if a single casualty causes a loss of control of a rudder, by effecting a change of power units at the Bridge Steering Unit or Aft Steering Unit (see 6.6), full control of the rudder will be regained within 4 seconds after initiating the power unit transfer. This shall apply to the hydraulic power units, including the motor controllers, and the complete control subsystem, including the rudder angle transmitter units and their drive linkages. This does not apply to a failure of the rudder actuator, actuator linkage, or actuator piping, nor to a failure of the hydraulic power unit transfer valve structural integrity.

3.2.1.6 Arrangements.

3.2.1.6.1 Component arrangement within compartment. Parts of the steering system, except power unit head tanks and bulkhead-mounted controls, shall be not less than 300 mm (12 inches) from the bulkheads and overhead deck framing of the steering gear room.

3.2.1.6.2 Guards and covers. Moving parts that are safety hazards shall be shielded with guards or covers that are easily removable for inspection purposes.

3.2.1.6.3 Securing of fasteners. Nuts, screws, keys, splines, and collars shall be secured to prevent their loosening. Set screws shall not be used.

3.2.1.6.4 Component separation. The hydraulic power units, local control units, and motor controllers of each steering gear shall be separated from the other corresponding units to the maximum extent practicable to enhance steering system survivability.

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3.2.2 Reliability and maintainability.

3.2.2.1 Reliability and maintainability requirements. The mean time between failure (MTBF) of each hydraulic power unit shall be not less than 8000 hours. The mean time to repair (MTTR) for all shipboard repairable equipment in the steering system shall be not greater than 1 hour. Not less than 90 percent of all shipboard repair actions shall be accomplished within 3 hours. The MTBF of steering system equipment that is not shipboard repairable shall be not less than 20,000 hours. Shipboard repairable shall be defined as corrective maintenance by forces afloat, performed in accordance with instructions included in the steering system technical manuals, using ship spares and tools. R&M requirements shall be imposed on all subcontractors (see 6.2 and 6.3).

3.2.2.1.1 Maintainability. The steering system machinery shall be designed for maintenance by no more than two persons, neither of whom shall require a skill level higher than first class engineman. Navy personnel skills shall be as specified in NAVPERS 18068. The steering gear controls shall be designed for maintenance by no more than two persons, neither of whom shall require a skill level higher than first class electrician (see 6.3).

3.2.2.1.2 Reliability predictions. Reliability predictions for the mechanical parts of the steering systems shall be based upon data from Fleet experience, when appropriate. When data is not available, mechanical part failure data may be obtained from the Rome Air Development Center (RADC), NPRD-2 "Nonelectronic Parts Reliability Data," or other valid source as approved by NAVSEA. Reliability predictions for the steering system controls shall be performed in accordance with MIL-HDBK-217 and shall be based on actual part stresses (calculated or measured).

3.2.2.1.3 Preventive maintenance. The minimum time between preventative maintenance (PM) services shall be 5 days or 24 operating hours, whichever occurs first. PM services for the steering system shall not require greater than 8 man-hours per month (see 6.3).

3.2.2.1.3.1 In-port maintenance. Not more than 1 hour of PM and operating time shall be required each 30-day period of nonuse to protect the system from deterioration.

3.2.3 Human performance and human engineering. Human engineering design criteria and principles used in the design of the steering system shall be in accordance with ASTM F 1166.

3.2.4 Interchangeability. Removable parts, subassemblies, and assemblies shall be interchangeable with corresponding items from a similar equipment, from stock or from production, without the necessity for reforming (for example, drilling, bending, filing), without the use of undue force, and without the necessity for individual selection. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

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3.2.5 Accessibility. Parts shall be as lightweight and compact as practicable, consistent with the required strength and stiffness, and shall be arranged to provide areas for inspection, repairs, replacement, and lubrication. Parts subject to periodic overhauling shall be removable without dismantling adjacent parts.

3.2.6 Shock, vibration, and noise.

3.2.6.1 Shock. Components of the steering system shall withstand grade A shock requirements of MIL-S-901. The yield strength of the steering gear materials shall not be exceeded under shock loads.

3.2.6.2 Vibration. The steering system shall be as immune to a vibrating environment as feasible. All rotating machinery shall be balanced to minimize vibration in accordance with MIL-STD-167-1. The steering system shall operate continuously and reliably during and after exposure to type I vibration, from 4 hertz (or lowest attainable frequency) up to and including 50 hertz (Hz), in accordance with MIL-STD-167-1 and shall withstand internally excited vibration in accordance with MIL-STD-167-1, type II.

3.2.6.2.1 Swaybraces. Where employed to afford stability under vibration, swaybraces shall be designed to fail or permit large deformations under a load caused by a force of five times the weight of the equipment applied at the center of gravity of the item.

3.2.6.3 Noise. When tested in accordance with 4.7.4.2.5, the steering system equipment shall meet the requirements specified herein. Steering system equipment located in the steering gear room shall not exceed the grade D airborne sound level acceptance criteria specified in MIL-STD-740-1. Pumps and valves shall not exceed the structureborne acceptance criteria specified in MIL-STD-740-2 for type II equipment. Other machinery components in the steering gear room shall not exceed the structureborne acceptance criteria for type III equipment.

3.2.7 Instruction charts. Safety instructions and electrical and mechanical operating instruction charts shall be provided for the steering system. Instruction charts shall be type F or H of MIL-P-15024. The charts shall be labeled OPERATING CHART for the equipment concerned. Charts shall be 380 mm (15 inches) high and not more than 910 mm (36 inches) long.

3.2.7.1 Hydraulic and electrical diagrams. An instruction chart shall be provided containing a hydraulic flow diagram and an electrical schematic diagram of the combined steering system machinery and controls. This chart shall be mounted in the steering gear room near the power units.

3.2.7.2 Operating instruction charts. Operating instruction charts shall be provided and mounted near the Aft Steering Unit in the steering gear room. The charts shall provide the information for the controls and switches used by the operator for start up, operation, transfer of power units, changing modes of control, and securing of equipment under normal and emergency conditions. The charts shall provide instructions for equipment checks between operators in the pilot house and in the steering gear room. The charts shall contain a complete

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check list of items that require checking and setup prior to startup of equipment, and a list of checks required during operation of equipment after startup. The information on all charts shall be arranged in logical order as follows:

(a) Equipment checks to make certain that maintenance functions required prior to startup have been performed.
(b) Setup and startup procedures followed by operating checks to determine the equipment will operate properly in all modes of operation.
(c) The procedure for setting of all controls to secure the equipment.

3.2.7.3 Lubrication charts. A lubrication chart shall be provided in the steering gear room. It shall show the equipment diagrammatically and shall contain complete instructions for its care and lubrication, including designation (Navy symbol) of the lubricants and hydraulic fluids required and the frequency of lubrication, analysis, filtration, or other maintenance required.

3.2.8 Transportability. The steering system equipment shall be transportable by rail, barge, truck, or ship without deterioration, degradation, or damage. General environmental considerations for transportation of the equipment shall be in accordance with SAE HIR 1063. Components of the steering system shall not be damaged by air transportation at altitudes up to 15.25 km (50,000 feet) in nonpressurized compartments.

3.2.9 Calculations and stress diagrams. (See 6.3.)

3.2.10 Maintenance and repair standards. (See 6.3.)

3.2.11 Purchase orders. (See 6.3.)

3.2.12 Allowance documentation. (See 6.3.)

3.3 Materials.

3.3.1 Material requirements. Materials shall meet all requirements specified herein and shall be free from defects which affect the serviceability of the steering system.

3.3.2 Structural design. The steering system shall be of the size and weight specified (see 6.2) without the combined stress at any point in mechanical and structural parts exceeding 35 percent of the part material yield strength, except where otherwise specified herein. Shock loads shall not be added to normal operational loads when computing maximum combined stress. In stress calculations, the following strength relationship shall be assumed to exist.

Elastic strength	Percent tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160

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3.3.3 Metals. Metals shall be compatible with the intended temperature, function, service, and storage conditions to which the components will be exposed. The metals shall possess corrosion-resistant characteristics or shall be protected by anodizing or the use of coatings to resist corrosion which may result from such conditions as dissimilar metal combinations, moisture, salt spray, and high temperature deterioration, as applicable. Low ductility material or high transition temperature material shall not be used in pressure containing boundaries of components. Low ductility material and high transition temperature material may be used in non-pressure boundary and non-load carrying applications where suitability is proven by high-impact shock tests. Low ductility material is metallic material showing less than 12 percent elongation in a standard tension test. High transition temperature material is material having a nil-ductility transition temperature (NDTT) above the minimum operational temperature. The NDTT shall be as measured by the drop-weight test in accordance with ASTM E 208. Charpy V-notch impact values may be used as criteria when these have been correlated with drop-weight test results.

3.3.4 Castings. Machinery castings shall conform to the requirements specified in 3.3.3 except that nondestructive testing, inspection, and acceptance shall be as specified in MIL-STD-278. Nondestructive testing, inspection, acceptance, and welding of structural castings shall be in accordance with MIL-STD-1689.

3.3.5 Welding fabrication. Welded fabrication, stress relieved, is acceptable in lieu of castings. A documented log of the stress-relieving cycle shall be maintained. A full description of welding procedures, using welding symbols conforming to MIL-STD-22, shall be maintained and made available to the Navy for review if requested. Welding for machinery components and piping shall be in accordance with MIL-STD-278. Welding for structural components shall be in accordance with MIL-STD-1689.

3.3.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Rudder actuator assemblies.

3.4.1 Assembly. A rudder actuator assembly shall comprise one or two ram assemblies, each working in a pair of opposed cylinders. The middle of the ram shall be suitably arranged for attaching the links of a type I steering system or the rapson slide mechanism of a type II steering system. Either one or two ram assemblies shall be provided for each actuator of type II steering systems as specified (see 6.2).

3.4.1.1 Structural integrity. Parts of the actuator assembly, including the foundation or bedplate, shall be of such strength and rigidity that no deflection, bending, or misalignment will occur to cause binding or excessive strain in any

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part when subjected to maximum load. Provision shall be made to allow the rudder stock to rise or drop at least 9.5 mm (0.375 inch) without subjecting any part of the actuator assembly to stresses greater than those encountered in normal operation.

3.4.1.2 Ram force efficiency. The efficiency used in converting ram force to rudder torque for calculating ram size shall be 80 percent for type I steering systems and 70 percent for type II steering systems.

3.4.1.3 Actuator lubrication. Arrangements for pressure grease lubrication of actuator assemblies shall be provided.

3.4.2 Cylinders. Cylinders shall be one-piece castings, forgings, or weldments employing 100-percent penetration welds. Cylinder heads shall be hemispherical. Supporting bosses or feet and lugs for glands and tie rods shall be integral with the cylinder walls. Cylinder bushings shall be provided, and the cylinder bore shall be sized so that the ram bears only on the bushings under all operating conditions up to and including maximum load and rudder angle. The bushing material shall be bronze in accordance with ASTM B 62, and the bearing pressure shall not exceed 14 MPa (2000 lb/in^2) over the projected area of the bushing at the maximum load on the actuator at any rudder angle.

3.4.2.1 Cylinder tie rods. The cylinders of each actuator assembly shall be connected to each other by one or more tie rods. The tie rod(s) shall be removable without moving the cylinders. One tie rod shall be used as a guide to prevent ram rotation, and shall be threaded and provided with manually operated jacking nuts for securing and positioning the rudder. The threaded tie rod and nut shall withstand the force developed when either ram cylinder is pressurized to 110 percent of MDWP, without exceeding 50 percent of the rod and nut material yield strength. One easily operated ratchet-type wrench shall be provided for each jacking nut on the actuator.

3.4.2.2 Cylinder packings. Cylinders shall be provided with standard "U" or "V" external packings in accordance with MIL-P-2911. Packings shall be accessible for easy replacement and adjustment. Wipers shall be installed ahead of the packings. The cylinders shall be arranged so that they can be slipped off the ram ends for maintenance with minimum disturbance to the surrounding equipment.

3.4.2.3 Cylinder piping connections. Each pipe connecting the transfer valve of each power unit to the actuator cylinders, and the cylinder drain pipes, shall be individually connected to the cylinders. Ball-type isolation valves with means for securing the valves in the open and closed positions shall be provided in all cylinder piping. The isolation valves shall be rated for at least 31 MPa (4500-lb/in^2) working pressure. The piping and fittings between the cylinders and the isolation valves shall be rated for at least 31 MPa (4500-lb/in^2) working pressure. Flanges, bosses, and attachments that are welded to the cylinders shall be of a material compatible with the cylinder material so that welding may be accomplished without the need for special welding procedures or oven preheat.

3.4.2.3.1 Piping requirements. The piping from the transfer valve of each power unit shall be connected on opposite sides of the cylinders near the cylinder ends. The cylinder drain ports shall be in the bottom of the cylinders. All cylinder isolation valve handles shall be nominally horizontal in the normal

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steering mode, and vertical in the emergency steering mode. The drain ports and connected piping shall be sized to accommodate the maximum flushing flow rate of the fill and drain pump. Cylinder vents shall be provided. Quick-disconnect fittings shall be provided in the drain piping downstream of the drain valves for connecting a fill and drain pump portable hose to fill and drain the cylinders.

3.4.3 Ram assemblies. The ram of each assembly shall be of a one-piece forging or weldment. The ram bearing surface shall be hardened to at least 50 Rockwell C, hard chrome-plated, and ground to at least a 0.25 to 0.38 micron (10 to 15 micro-inch) finish. The chrome plate minimum thickness shall be 0.05 mm (0.002 inch) after grinding.

3.4.3.1 Ram clearances. The ram shall clear the ends of the cylinders by at least 12.7 mm (1/2 inch) when the ram crosshead is on the steel stops. The crosshead sliding blocks of rapson-slide-type actuating linkage shall not reach within 6.35 mm (1/4 inch) of the outboard ends of the crosshead slots when the ram is on the steel stops. When the rudder is amidships, the sliding blocks of rapson-slide-type actuating linkage shall not reach within 6.35 mm (1/4 inch) of the inboard ends of the crosshead slots.

3.4.4 Ram overtravel stops. Steel stops shall be provided and shall positively limit the rudder travel to 3 degrees beyond the specified maximum working angle. Crushable copper pieces shall be provided ahead of the steel stops and shall cushion the blow if the rudder enters the stops at the maximum design rate and maximum load. The copper pieces shall be designed to be engaged 1 degree before the steel stops. The copper pieces shall remain effective after the rudder has hit the stops at least twice at the maximum design rudder rate and maximum load. Where two or more stops limit the rudder travel, engagement of the stops shall occur consecutively within 0.3 mm (0.01 inch).

3.4.5 Indicator. A mechanical rudder angle indicator shall be provided on the rudder actuator or rudder stock. The indicator shall be accurately graduated in 1-degree increments with numbers at each 5-degree mark. The indicator shall be clearly visible from the emergency steering control valves on the fill and drain pump unit (see 3.5.4.1) and from the location of the tie rod jacking nuts. Two mechanical indicators shall be provided for each actuator, if necessary, for visibility from the two locations.

3.5 Hydraulic system.

3.5.1 General.

3.5.1.1 Pressure rating. High pressure components shall be rated for at least 18 MPa (2600 lb/in^2) working pressure, except where specified otherwise herein.

3.5.1.2 Pipe sizing. Piping shall be sized to limit fluid velocity to no greater than 6100 mm/s (20 feet per second) and to meet the pressure loss requirements specified in 3.5.2.8.1.5. Suction piping and fittings shall be sized to provide adequate pressure at the pump inlet to prevent audible pump cavitation when the hydraulic fluid temperature is 16°C (60°F) and higher, and the head tank is at the low level mark.

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3.5.1.3 Piping material and hydraulic components. Hydraulic components shall be in accordance with MIL-STD-2193. Pipe material selection shall be in accordance with MIL-STD-777. Pipe fitting connections shall be welded. If tubing is used for pressure gages, it shall be corrosion resistant steel (CRES). Piping shall be CRES.

3.5.1.3.1 Pump piping. Pump piping shall be arranged so that the pump may be removed without breaking pipe connections other than at the pump.

3.5.1.3.2 Hoses. Hoses shall be in accordance with MIL-H-24135. Hose material shall be compatible with the hydraulic fluid (see 3.5.1.4). Where practicable, 90-degree dogleg hose assemblies shall be used to connect solid-mounted equipment to resiliently mounted equipment. The hose configuration and length shall be sufficient to accommodate the motion between equipment allowed by the mounts, without stressing the hose assembly or associated fittings and machinery. Hoses shall not be subjected to torsional or twisting stresses and shall remain clear of surrounding equipment when moving through maximum excursion.

3.5.1.4 Hydraulic fluid. Hydraulic fluid shall be in accordance with MIL-H-17672, type 2135.

3.5.1.4.1 Hydraulic fluid temperature. Normal operating temperature range for the hydraulic fluid shall be from 16°C (60°F) to 57°C (135°F).

3.5.1.5 Pressure gages. Pressure gages shall be provided as specified in 3.5.2.1.3.1 and 3.5.2.1.4.4 and shall be in accordance with MIL-I-18997. Snubbers and isolation valves shall be provided for each pressure gage. Power unit pressure gages shall be in clear view of an operator at the Aft Steering Unit.

3.5.1.6 Temperature indicators. Temperature indicators shall be provided as specified in 3.5.2.4.4 and shall be in accordance with MIL-I-17244. Thermowells shall be selected and installed in accordance with Drawing 810-1385917.

3.5.1.7 Fluid filters. Hydraulic fluid filters shall be provided as specified in 3.5.2.5 and 3.5.4.3. All filter assemblies provided for the steering system shall comprise identical, disposable-type filter elements. Each filter assembly shall comprise a bypass relief valve and a filter element pressure drop indicator that is in clear view of ship's watch personnel. The filter assemblies shall be designed to minimize fluid spillage during element replacement and shall be equipped with vent and drain plugs. The filter assembly installation shall be designed such that element replacement can be accomplished without draining any other components.

3.5.1.7.1 Filter elements. The filter elements shall be in accordance with MIL-F-24702 and MIL-F-24702/1 and shall have a Beta 10 ratio of 75 minimum.

3.5.1.7.2 Filter condition sensing switch. A switch shall be provided for each power unit filter that shall activate the MINOR FAULT light on the power unit motor controller and Bridge and Aft Steering Units when the filter element requires replacement. The switch shall actuate at a pressure differential that is less than the bypass relief valve cracking pressure. The fault light shall not be actuated by transient pressures. The switch shall be electrically disabled when the supply tank temperature is 38°C (100°F) and below.

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3.5.1.8 Provisions for fluid sampling. Vent and test valves in accordance with MIL-V-24695 shall be provided at various locations throughout the hydraulic system where fluid contamination and pressure level sampling is required for maintenance purposes. The sample fittings shall also be used where vents are required and shall be provided in manifolds and piping at suitable locations for sampling the differential pressure across all major hydraulic components.

3.5.2 Power unit. Two power units shall be provided for each machinery subsystem. Each power unit shall be provided with a dedicated local control unit and motor controller that interface with the steering control subsystem. Each power unit shall be provided with a remotely operated transfer valve. The valve shall connect the power unit to the rudder actuator in the active (RUDDER ENGAGED) mode and shall bypass the pump in the standby mode and when starting the pump motor. Each power unit shall be capable of operating continuously in either the active mode or in the standby mode when the pump stroke is no greater than 25 percent maximum stroke. Means shall be provided to start and stop each power unit from the remote steering control units and from the power unit motor controller.

3.5.2.1 Main pump. The main hydraulic pump in each power unit shall be a reversible flow, variable delivery, piston type in accordance with MIL-P-17869. The pump shall be driven at a speed not greater than 50 percent of the maximum rated drive speed for the MDWP and shall have a continuous operating pressure rating at least twice the MDWP.

3.5.2.1.1 Servo valve control. The pump displacement (stroke) and flow direction shall be controlled by a manifold-mounted, 2-stage, electrohydraulic servo valve operating in conjunction with a mechanically linked, pump stroke feedback, dual potentiometer assembly in accordance with MIL-R-39023 and the power unit's dedicated local control unit. The flow from the electrohydraulic servo valve shall act directly on the pump stroke actuators. The servo valve torque motor shall be provided with dual coils. The potentionmeters shall be easily replaceable and shall be easily nulled by ship maintenance personnel. Only one coil shall operate the servo valve at any one time in all steering modes. The steering control subsystem shall select which coil is active. There shall be no failure mode in the control subsystem, excluding loss of all ship electrical supply power or loss of the hydraulic power unit helm order signal, that results in simultaneous loss of both coils in one servo valve.

3.5.2.1.2 Pump displacement indicator. A mechanical indicator shall be located on the pump to show the percentage and direction of pump flow.

3.5.2.1.3 Pump pressure protection. The pump shall incorporate externally adjustable crossblow relief valve protection and a hydraulically powered override of the electrohydraulic stroke control when the pump pressure is at 110-percent MDWP and higher. The override shall be implemented without electrical control and shall be capable of reducing the pump stroke from maximum to nominal zero in less than 0.1 second with the servo valve wide open in the direction that opposes the override. The override shall sustain the pump steady-state pressure at 110 ± 5-percent MDWP, if the rudder impacts the hardstops and the control system continues to call for fluid flow in that direction.

3.5.2.1.3.1 Pressure gauges for main pump output ports. Pressure gages shall be provided for each power unit and shall continuously monitor the pressure in both output ports of the main pump.

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3.5.2.1.3.2 Pump case filling and draining connection. A quick disconnect fitting shall be provided in the bottom of the pump case for connecting a fill and drain pump portable hose for filling and draining the pump case.

3.5.2.1.4 Auxiliary pumps. The main pump shall comprise one or two auxiliary pumps to furnish servo and replenishment fluid. The sustained replenishment flow rate shall be at least 12 percent of the flow rate required from the main pump to develop the maximum design rudder rate. The sustained servo flow rate shall be sufficient for the servo valve to move the main pump swashplate from zero to maximum in 0.5 second or less. Pilot-operated relief valves shall be provided and shall control servo and replenishment pump pressures.

3.5.2.1.4.1 Auxiliary pump arrangement. The auxiliary pumps shall be directly driven by the main pump drive shaft. The auxiliary pumps shall be easily replaceable without moving the main pump on its support. One auxiliary pump may be used to supply both servo and replenishment requirements if the flow capacity is sufficient for both.

3.5.2.1.4.2 Replenishment check valves. The replenishment fluid circuit shall connect with the main pump output piping via spring-loaded check valves of adequate size, to prevent audible main pump cavitation under all normal steering operations.

3.5.2.1.4.3 Servo pump discharge filter. A filter assembly shall be provided in the servo pump discharge and shall filter all the fluid going to the servo valve and to the transfer valve pilot control valve.

3.5.2.1.4.4 Auxiliary pump pressure monitoring. A pressure gage shall be provided to continuously monitor the pressure between the servo filter and the servo valve inlet. A pressure gage shall be provided to continuously monitor the replenishment pump pressure. A pressure switch shall be provided to activate the MAJOR FAULT light on the motor controller and the Bridge and Aft Steering Units when the servo pressure is low. This switch shall be disabled during power unit startup and normal shutdown, and shall not be activated by pressure transients.

3.5.2.1.4.5 Case pressure protection. A spring-loaded check valve shall be provided immediately adjacent to the pump case drain port. The flow through the check vale shall lead directly to the supply tank above the tank pump suction strainer specified herein. It shall be sized and set to protect the pump case against excessive pressure due to cold temperatures and return flow surges. The valve shall not pass any flow when the flow through the pump case is equal to the full flow of the auxiliary pump(s) and the hydraulic fluid temperature is 38°C (100°F) and higher.

3.5.2.2 Replenishment accumulator. A gas-charged, bladder-type accumulator shall be provided to supplement the replenishment pump. The rated size of the accumulator shall be at least 2.5 percent of the fluid volume in the rudder actuator assembly. Engineering calculations shall be prepared that show that the accumulator piping arrangement provides sufficient replenishment flow to prevent audible main pump cavitation under all normal steering conditions.

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3.5.2.2.1 Accumulator piping for servo pressure backup. When separate servo and replenishment pumps are provided, check valves shall be provided and shall allow the replenishment accumulator to supply fluid to the transfer valve pilot valve when servo pressure becomes less than the accumulator pressure.

3.5.2.2.2 Accumulator pressure switch. A pressure switch shall be provided and installed immediately adjacent to the replenishment accumulator. The switch shall activate the MAJOR FAULT light on the power unit motor controller and the Bridge and Aft Steering Units when the replenishment pressure becomes equal to or less than the initial accumulator gas precharge pressure. The pressure switch shall be electrically disabled when the power plant is not operating and during power unit startup.

3.5.2.3 Main pump motor and controller. One electric motor shall be provided to drive each main pump. The power sources for each steering gear shall be arranged so that a single failure will not result in loss of power to both hydraulic power units. The motor shall be in accordance with MIL-M-17060 and the following:

Service: Navy A
Ambient temperature: 50°C (122°F)
Voltage: 440 volts alternate current (Vac), 60 Hz, 3-phase
Winding: squirrel cage, induction
Duty: as required
Horsepower: not to exceed 150 percent of the nameplate rating at any point on the rudder torque curve
Design: B
Enclosure: totally enclosed

3.5.2.3.1 Electric motor stall. The motor shall not stall if the rudder hits the hardstops and the pump displacement is the maximum allowed by the pump stroke mechanical stops.

3.5.2.3.1.1 Motor overload and overheating protection. Switching-type thermistors shall be imbedded in the winding of each motor. Three thermistors shall be provided for motor overload protection and shall perform this function whenever the motor is running. Three thermistors shall also be provided to activate the MAJOR FAULT light on the motor controller and the Bridge and Aft Steering Units to provide advance warning of high motor temperature. The outputs of these thermistors shall be processed by the steering control subsystem.

3.5.2.3.1.2 Motor insulation. Each motor shall be provided with a sealed insulation system.

3.5.2.3.2 Motor controllers. Each power unit shall be provided with a dedicated motor controller. Each motor controller shall provide for local and remote control of the power unit main pump motor and transfer valve, provide supply power for the power unit local control unit, and provide status information to the local operator and to the steering control subsystem. Each motor controller shall be in accordance with MIL-C-2212 and the requirements herein.

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Control voltage - 115 Vac, 60 Hz
Enclosure - drip-proof
Operation - magnetic across the line
Duty - as required
Protective features - Overload protection shall be provided. Low-voltage-release (LVR) shall be provided when the power unit is active, and low-voltage-protection (LVP) shall be provided when the power unit is in standby. When both power units are operating the rudder simultaneously, LVR shall be applied to the first power unit motor running with its transfer valve activated; LVR shall be applied to the last power unit running with its transfer valve activated when both power units are shut down. These protective features shall be incorporated in the local control unit.

3.5.2.3.2.1 Sensor inputs. The main contactor control relay shall be controlled by the power unit high high temperature, low low level, and motor overload sensors such that activation of any of these sensors shall automatically shut down the power unit and activate the CRITICAL FAULT light on the motor controller and the Bridge and Aft Steering Units. The fault light shall be provided with an identification label with the inscription CRITICAL FAULT.

3.5.2.3.2.1.1 Loss of 440-Vac power. Loss of 440-Vac power to the motor controller shall result in a critical fault indication. A time delay relay shall be provided to prevent automatic shutdown and transfer to the standby power unit when the power loss lasts less than 2 seconds except for power failure.

3.5.2.3.2.1.2 Power unit transfer. Each of the two power units (including the associated motor controllers and local control units), that operate one rudder shall be interconnected so that stopping of the active power unit for any reason shall cause:

(a) The standby unit to start up automatically, if it is not already running.
(b) The transfer valve to activate after the delay specified herein.
(c) LVR protection to be transferred to the standby power unit, which shall then become the active power unit.

3.5.2.3.2.2 Motor controller transformer. Each motor controller shall comprise a 440/115-Vac transformer for supplying control power for the motor controller and for activating the power unit transfer valve. Power for activating the transfer valve shall be switched off by an auxiliary contact on the main contactor when the main contactor is not energized. The motor controller shall also provide ac supply power for the power unit dedicated local control unit.

3.5.2.3.2.3 Switches for motor controller. Each motor controller shall comprise the following switches on the front panel of the controller: a local START MOTOR switch, a local STOP MOTOR switch, a MODE SELECTOR switch, and a local ENGAGE RUDDER switch. The switches shall be arranged as generally shown on figure 1.

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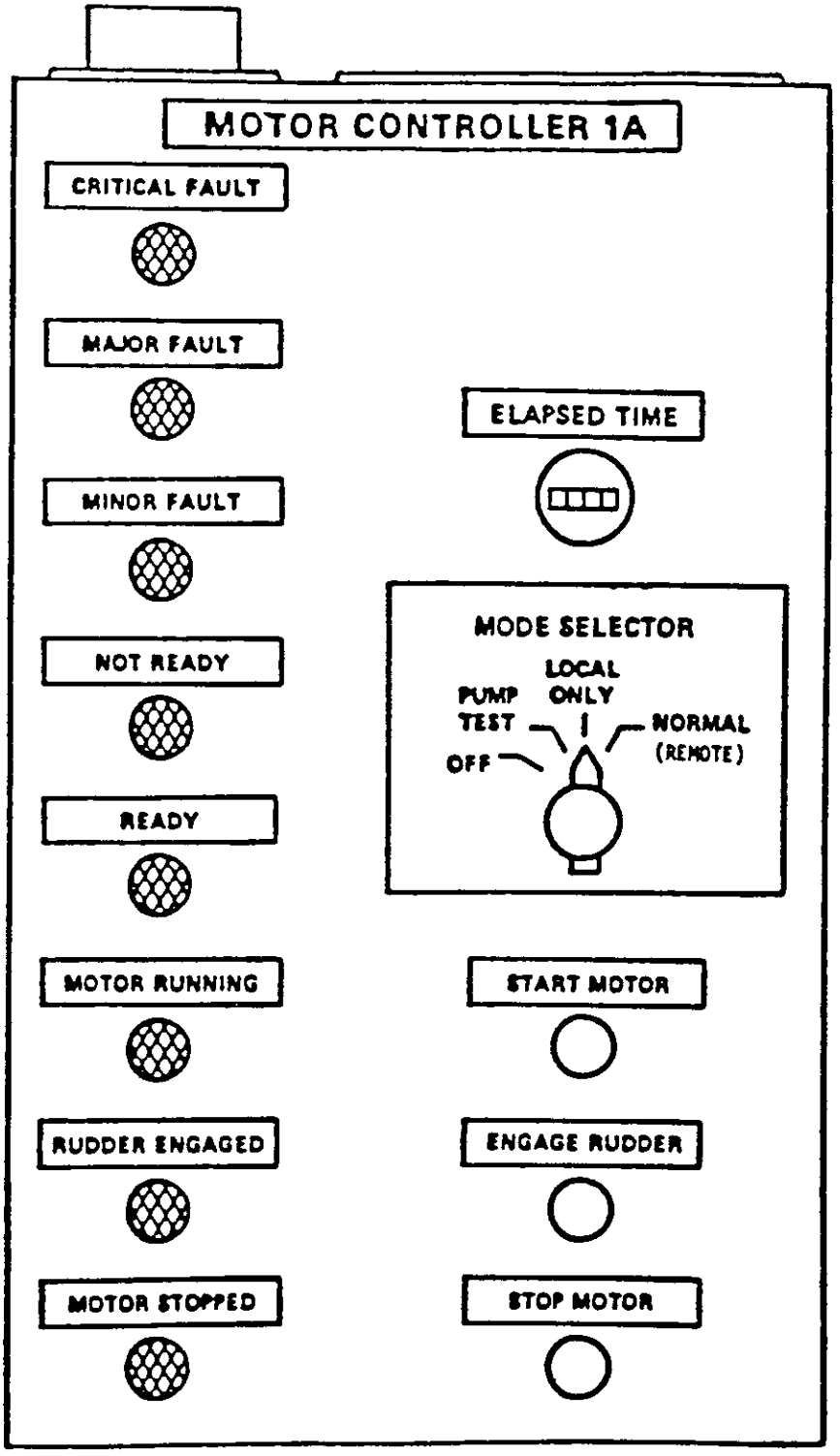


FIGURE 1. Motor controller front panel.

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3.5.2.3.2.3.1 MODE SELECTOR switch. The MODE SELECTOR switch shall be a rotary, 4-position, multideck tap switch in accordance with MIL-S-18396. Switch contacts shall be provided for indicating switch position to the steering control subsystem.

3.5.2.3.2.3.1.1 The function of the MODE SELECTOR switch. The MODE SELECTOR switch shall be provided with the following positions: NORMAL (REMOTE), LOCAL ONLY, PUMP TEST, and OFF. The switches and MODE SELECTOR switch positions shall be identified on the face of the motor controller, as indicated by the capitalized nomenclature used herein. The MODE SELECTOR switch shall provide the following:

(a) Switch off the high and low sides of the 115-Vac voltage supply from the 440/115-Vac transformer in the OFF position.
(b) Switch off the high and low sides of the transfer valve connection in the OFF position.
(c) Allow the local ENGAGE RUDDER switch to function in the NORMAL (REMOTE) and LOCAL ONLY positions only.
(d) Allow remote START MOTOR and STOP MOTOR switches and remote ENGAGE RUDDER switches to function in the NORMAL (REMOTE) position only.
(e) Allow the motor to be started when the local START MOTOR switch is depressed, but switch off the high and low sides of the transfer valve connection in the PUMP TEST position.

3.5.2.3.2.3.1.2 Interconnecting motor controllers and local control units. Each of the two motor controllers and associated local control units shall be interconnected so that:

(a) Switching the MODE SELECTOR switch from the OFF position to either the LOCAL ONLY or NORMAL (REMOTE) position shall start the power unit motor, activate the power unit transfer valve after a suitable time delay to allow the motor to come up to normal operating speed, and assign LVR protection to the activated power unit when the other power unit MODE SELECTOR switch is in either the OFF or PUMP TEST positions.
(b) Switching the MODE SELECTOR switch from the OFF position to the PUMP TEST position shall permit the power unit motor to start, but shall never activate the power unit transfer valve when the standby power unit MODE SELECTOR switch is in the OFF or PUMP TEST positions.

3.5.2.3.2.3.2 Push button switches. The local START MOTOR, STOP MOTOR, and ENGAGE RUDDER switches shall be push button, momentary contact type.

3.5.2.3.2.3.2.1 START MOTOR switch. Depressing the standby power unit local START MOTOR switch shall only start the power unit motor. If there is a critical fault in the power unit, the motor shall not start.

3.5.2.3.2.3.2.2 ENGAGE RUDDER switch. Depressing the standby power unit local ENGAGE RUDDER switch shall start the standby power unit motor, if it is not already running, activate the power transfer valve after a time delay relay has allowed sufficient time for the motor to come up to normal operating speed, and transfer LVR protection. If the standby motor is already running, depressing the

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standby power unit ENGAGE RUDDER switch shall immediately activate the standby power unit transfer valve and transfer LVR protection. If there is a critical fault in the standby unit, the active power unit shall not shut down if the standby power unit ENGAGE RUDDER switch is depressed.

3.5.2.3.2.4 Roll stabilization mode. If the roll stabilization option is required (see 6.2), it shall be possible to engage the standby power unit without disengaging the active power unit. In that mode only, the standby power unit shall be started (if not already running) and engaged in parallel with the active power unit simply by pressing its ENGAGE RUDDER switch. With this arrangement, both power units shall have LVR while they are running. However if either loses power, even momentarily, it shall revert to LVP and shall stay off the line.

3.5.2.3.2.5 Motor controller supplied information. The following information shall be supplied from each motor controller to the steering control subsystem:

(a) When ac power is available at the motor controller.
(b) When the transfer valve is in the activated (RUDDER ENGAGED) position.
(c) When there is a CRITICAL FAULT, MAJOR FAULT, or MINOR FAULT indicated on the motor controller panel as specified herein.
(d) When the main contactor is closed.
(e) When the transfer valve control relay is activated.
(f) When the LVR control relay is latched.
(g) The state of all time delay relays.
(h) The position of the MODE SELECTOR switch.

3.5.2.3.2.6 Indicators. Each motor controller shall comprise the following indicator lights on the front face of the controller: MOTOR RUNNING, MOTOR STOPPED, READY, NOT READY, CRITICAL FAULT, RUDDER ENGAGED, MAJOR FAULT, and MINOR FAULT. Indicator lights shall be arranged as generally shown on figure 1 and shall be identified on the face of each motor controller by the capitalized nomenclature used herein.

3.5.2.3.2.6.1 MOTOR RUNNING indicator. The MOTOR RUNNING light shall activate only when the main contactor is energized.

3.5.2.3.2.6.2 MOTOR STOPPED indicator. The MOTOR STOPPED light shall activate only when the main contactor is not energized.

3.5.2.3.2.6.3 NOT READY indicator. The NOT READY light shall be activated only when the READY light is not activated.

3.5.2.3.2.6.4 READY indicator. The READY light shall activate only when all of the following are true for the power unit:

(a) 440-Vac power is available at the motor controller.
(b) At least one of the two power supplies within the power unit's local control unit is functioning properly.
(c) The switches in the associated local control unit are in their NORMAL operating positions.

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(d) The MODE SELECTOR SWITCH is in the NORMAL (REMOTE) or LOCAL ONLY operating position.
(e) No critical fault is indicated.

3.5.2.3.2.6.5 CRITICAL FAULT and RUDDER ENGAGED indicators. The CRITICAL FAULT and RUDDER ENGAGED lights shall be activated as specified herein.

3.5.2.3.2.6.6 MAJOR FAULT indicator. The MAJOR FAULT light shall be activated on the motor controller and the Bridge and Aft Steering Units when any of the following are true for the power unit.

(a) The low replenishment pressure switch is activated.
(b) The low servo pressure switch is activated.
(c) A servo control fault occurs.
(d) The low level switch is activated.
(e) The high oil temperature switch is activated.
(f) High motor temperature is detected by imbedded temperature sensors provided in the motor as specified herein.

3.5.2.3.2.6.7 MINOR FAULT indicator. The MINOR FAULT light on the motor controller and the Bridge and Aft Steering Units shall be activated when any of the following are true for the power unit:

(a) The return oil filter high pressure switch is activated.
(b) The servo oil filter high pressure switch is activated.
(c) Loss of the power unit local control unit redundant channel or power supply.
(d) Other faults, as required to meet maintainability requirements specified herein.

The MINOR FAULT light shall be provided with an identification label with the inscription MINOR FAULT.

3.5.2.3.2.6.8 Indicator light transformer. All lights on the motor controller shall be powered by a 115/12-Vac transformer that shall be provided within each motor controller. Relays shall be provided for individually activating the lights. The relays shall be operated by the steering control subsystem.

3.5.2.3.2.7 Elapsed time meter. Each motor controller shall be provided with an elapsed time meter capable of displaying up to at least 9999 hours. The meters shall be activated when the main contactor switch is closed, and shall be located on the front face of the motor controller.

3.5.2.4 Cooler. A hydraulic fluid cooler shall be provided in the fluid circuit between the pump case drain and the supply tank inlet of each power unit. The cooler shall be a double-tube, sheet-type in accordance with MIL-C-15730, type A, class 3, except that double-tube construction is not required. The cooling capacity shall be sufficient to maintain the cooler inlet temperature at 57°C (135°F), or less, when the main pump is operating continuously at 25-percent MDWP average and 0 to 5-percent flow rate.

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3.5.2.4.1 Cooling water supply. The coolers for each steering gear shall be supplied cooling water from separate supplies in the ship, so that loss of a single cooling water supply will not result in loss of both power units connected to one rudder. Cooling water flow rate for each cooler shall be controlled by a fixed-flow control orifice that shall be fabricated from monel material and shall be installed in the cooling water return line. The piping immediately downstream of the flow control orifice shall be straight for at least 10 pipe diameters. Isolation valves shall be installed in the cooling water inlet and return lines.

3.5.2.4.1.1 Cooling water flow velocity. Cooling requirements shall be met with the cooling water velocity no greater than 1520 mm/s (5.0 feet per second) within the cooler. The cooler water velocity within the cooler shall not exceed 2000 mm/s (6.5 feet per second) when the cooling water supply is at its maximum design pressure.

3.5.2.4.1.2 Cooling water flow measurement provisions. Means shall be provided to positively indicate that the cooling water is flowing through the cooler.

3.5.2.4.1.3 Air entry protection. A trap or equivalent pipe loop shall be installed downstream of cooler so that air cannot enter the cooler through the cooling water return piping.

3.5.2.4.2 Cooler vent lines. Vent lines from the voids between the cooler double-tube sheet shall be arranged to run any leakage into a funnel that shall be in clear view of operating personnel.

3.5.2.4.3 Thermostatic control valve. The hydraulic fluid flow to the cooler shall be arranged to pass through a direct-acting, thermostatically operated, proportional control valve that shall enable all of the following:

(a) Block off the cooler flow and bypass the cooler.
(b) Pass all hydraulic fluid through the cooler.
(c) Partially bypass the cooler, as necessary to provide fluid temperature regulation. There shall be no valve operating position that blocks or partially blocks the fluid flow from the pump case drain.

3.5.2.4.4 Temperature indicator. A fluid temperature indicator shall be provided in the fluid circuit near the main pump case drain outlet that shall be easily visible by personnel.

3.5.2.4.5 High high temperature sensor. A switch-type thermistor shall be provided in the main pump case drain piping immediately adjacent to the pump, and shall activate the CRITICAL FAULT light on the power unit motor controller and the Bridge and Aft Steering Units when the case drain fluid temperature is 85°C (185°F) and above, and shall be arranged to automatically shut the power unit down.

3.5.2.5 Return filter. A filter assembly shall be provided for each main pump and shall filter all the fluid entering the supply tank, including fluid from the pump case drain and fluid being supplied by the fill and drain pump unit.

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3.5.2.6 Supply tank. A supply tank shall be provided for each power unit that shall be sized for a dwell time not exceeding 1.5 minutes. The supply tank and all its parts shall be fabricated from stainless steel. The tank interior shall be smooth and uniform to minimize sediment accumulation. The tank bottom shall be of conical shape, with not greater than 90 degrees included angle, and the tank suction port shall be at the bottom of the tank at the apex of the cone.

3.5.2.6.1 Supply tank strainer. The supply tank shall comprise a removable, 200-mesh, auxiliary pump suction strainer. The strainer shall withstand at least 0.1 MPa (15-lb/in^2) pressure differential in the normal flow direction. The strainer and tank assembly shall be arranged to continuously scavenge entrained air from the fluid and vent it to the head tank. The vent piping shall be arranged to prevent return fluid from bypassing the strainer. The vent piping shall not have low points that cannot drain back to the supply or head tanks. The tank and strainer air removal capability shall be equal to or better than a tank with dwell time of 4 minutes when the fluid is at and above the normal operating temperature.

3.5.2.6.1.1 Strainer pressure protection. A spring loaded check valve shall be provided to protect the strainer. The check valve cracking differential pressure shall be at least 0.05 MPa (7 lb/in^2), and the valve shall be sized so that it will pass the full of the auxiliary pump(s) without exceeding 0.1 MPa (15 lb/in^2) differential.

3.5.2.6.2 High temperature sensor. A thermistor-type temperature sensor shall be provided in the supply tank. The sensor output shall be processed by the steering control subsystem to actuate the MAJOR FAULT light in the power unit motor controller and the Bridge and Aft Steering Units when the fluid temperature exceeds 60°C (140°F). The sensor output shall be processed by the steering control subsystem to also disable the hydraulic filter high pressure switches when the fluid temperature is less than 38°C (100°F).

3.5.2.6.3 Supply tank design pressure. The supply tank shall be designed to withstand the maximum pressures of its intended use under the operating conditions specified herein.

3.5.2.6.4 Supply tank draining provisions. A quick disconnect fitting shall be provided in the supply tank suction piping for connecting a fill and drain pump portable hose for draining the tank.

3.5.2.6.5 Multiple supply tanks. More than one supply tank may be provided for each main pump, if necessary, to suit steering gear room space and machinery accessibility requirements. Multiple supply tanks for one pump shall be connected in parallel and arranged fore and aft at the same height. Common vent and drain arrangements for multiple supply tanks may be provided. Inlet and suction piping shall be arranged to ensure that all multiple supply tanks function equally.

3.5.2.7 Head tank. A head tank shall be provided for each power unit and shall be fabricated from stainless steel. The tank shall contain at least 6 gallons of hydraulic fluid, or at least 25 percent of the supply tank volume, whichever is greater.

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3.5.2.7.1 Head tank location and piping. The head tanks shall be located in the overhead of the steering gear room and supported by the overhead deck. The head tank bottom outlet shall connect to the supply tank outlet. The outlet shall connect to the power unit auxiliary pump suction piping, as close as practicable to the pump inlet. The connecting pipe size shall be sufficient to prevent auxiliary pump audible cavitation during power unit startup.

3.5.2.7.2 Head tank fluid level sight gage. The head tank shall be provided with a tank level sight gage, with low and full level marks provided. The low mark shall be at 1/2 of the head tank fluid volume. The head tank shall be located so that the tank level gage and the low and full marks are easily visible to personnel operating the system.

3.5.2.7.2.1 Head tank filling provisions. A fill line leading from the fill and drain pump manifold to the head tank via the supply tank piping shall be provided. The fill line connection to the power unit fluid circuit shall be in the piping between the main pump case drain port and the return filter. Each fill line shall comprise an isolation valve at the fill and drain manifold, and a fill stop valve that is located near the head tank so that an operator can see the level sight gage. No other power unit filling connections shall be provided.

3.5.2.7.3 Head tank cross connect. Head tanks for one machinery subsystem shall be located adjacent to and horizontally across from each other. One-inch nominal pipe size (nps) horizontal cross-connect piping shall connect the head tanks together. The cross-connect piping connections shall be made at or above the full level of each tank.

3.5.2.7.4 Head tank overflow pipe. The head tank cross-connect piping shall include a tee with the branch connection rising vertically and connecting to a head tank overflow pipe that shall lead to the top of the storage tank. The overflow pipe shall be sized to accommodate the tank fill rate at 16°C (60°F), and the maximum fill and drain pump flow rate at 54°C (130°F), without causing the head tank maximum working pressure to be exceeded. The overflow pipe shall not have low points that will not drain to the head or storage tanks.

3.5.2.7.5 Low level and low low level sensors. A switch shall be provided in the head tank and shall be set to actuate when fluid level is at and below the low level mark on the head tank level gage. The switch output shall be processed by the steering control subsystem to actuate the MAJOR FAULT light on the power unit motor controller and the Bridge and Aft Steering Units. A second switch shall be provided and arranged to automatically shut the power plant down and light the CRITICAL FAULT light on the power unit motor controller and the Bridge and Aft Steering Units if the head tank becomes empty. The shutdown switch shall not be activated due to ship motions specified in 3.2.1.4 when the tank is at the low fluid level.

3.5.2.7.6 Flexible connection between head tank and supply tank. A hose shall be installed in the piping connection between the head tank and supply tanks of each power unit and shall be of sufficient length to accommodate the maximum relative deflection between the tanks under shock conditions, without the connecting hose or piping parting.

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3.5.2.8 Transfer valve and manifold. A valve manifold shall be provided for each power unit between the power unit and the rudder actuator that shall comprise a transfer valve and a manually operated, cartridge or subplate-mounted-type rudder actuator bypass valve. Means shall be provided to secure the bypass valve in the closed position. The bypass valve shall be at least 1-inch nps.

3.5.2.8.1 Transfer valve function. The transfer valve shall be a manifold-mounted, solenoid pilot-valve-operated type. With the solenoid deactivated, the valve shall cause the main pump fluid to bypass and shall block the lines going to the rudder actuator. With the solenoid activated to put the power unit on-line, the valve shall connect the main pump output piping to the rudder actuator assembly. The valve shall not block fluid flow from the pump while shifting between the two positions.

3.5.2.8.1.1 Electric power for solenoid. The solenoid pilot valve shall use 115-Vac electrical power, which shall be provided by the main pump motor controller.

3.5.2.8.1.2 Pilot valve operation. The solenoid pilot valve shall be supplied fluid pressure from the servo pump and the replenishment accumulator. The transfer valve shall be capable of being shifted when the main pump pressure is the MDWP and the pilot pressure is not greater than replenishment pressure. The pilot valve shall have provisions for manual override to the activated and deactivated positions.

3.5.2.8.1.3 RUDDER ENGAGED pressure switch. A pressure switch shall be provided and shall be actuated when the pilot valve applies pressure to move the transfer valve to the active power unit position. The switch shall actuate RUDDER ENGAGED indicator lights on the Bridge Steering Unit and Aft Steering Unit and the power unit motor controller.

3.5.2.8.1.4 Transfer valve shift time. The transfer valve shall be shifted from the activated to the deactivated position in sufficient time to:

(a) Prevent the main pump motor from exceeding 150-percent design speed after the main pump motor loses electrical power while the rudder torque is the maximum aiding torque.
(b) Prevent the main pump motor from reversing direction after the main pump motor loses electrical power while the rudder torque is the maximum restoring torque.
(c) Meet the 4-second single casualty recovery time.

3.5.2.8.1.5 Main piping pressure drop. The total fluid pressure drop (supply and return) caused by the transfer valve, the manifold, and the interconnecting piping between the power unit and rudder actuator shall not exceed 1.4 MPa (200 lb/in^2) when the pump is on full stroke and the fluid is at or above the normal operating temperature.

3.5.2.9 Resilient mounts. Power unit equipment, except the head tank and motor controller, shall be mounted on a bedplate that shall be resiliently mounted using soundmounts in accordance with MIL-M-17508. Each power unit shall meet the airborne and structureborne noise requirements specified in 3.2.6.3.

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3.5.2.9.1 Power unit sub-bedplate. The main pump and motor shall be mounted on a sub-bedplate of such strength and rigidity as to maintain proper alignment between the pump and motor. The pump and motor unit and the sub-bedplate shall be resiliently mounted on the power unit bedplate.

3.5.2.9.2 Resonators. Sidebranch resonators tuned to the nominal fundamental pump frequency (revolutions per second times number of pistons) shall be provided and shall reduce the main pump fluid pressure pulsation noise before the fluid enters the rudder actuator. The resonators shall be 1/4-wavelength long, based on a sonic velocity of 1220 m/s (4000 feet per second), and shall be connected to the main pump output piping flanges with an orientation that allows self-venting to the main pump piping. The resonators shall be fabricated using the same size pipe as used for the main pump output piping.

3.5.3 Storage tank. One storage tank shall be provided for each steering gear room. Each tank at the nominal full level shall hold a fluid volume equal to twice the entire fluid volume in one power unit and one rudder actuator assembly. This entire volume shall be available for filling the system when the ship is permanently inclined, as specified in 3.2.1.4. In addition, the tank shall provide extra, normally empty, volume sufficient to receive the fluid from one power unit's supply and head tanks.

3.5.3.1 Storage tank elevation. The storage tank shall be elevated as necessary to prevent fill and drain pump audible cavitation when the fluid in the tank is no more than 1/2 full.

3.5.3.2 Storage tank filling provisions. The storage tank shall have an easily accessible 2-inch nps fill port and cap. A tank level sight gage shall be provided. A warning sign shall be provided that is highly visible to the operators and that cautions against filling the storage tank above the nominal full level.

3.5.3.3 Storage tank air breather filter. The storage tank shall be provided with a 10-micrometer nominal air breather filter. The breather shall be equipped with a dual air relief to maintain the air pressure in the storage tank between 0.002 MPa (0.3 lb/in^2) vacuum and 0.02 MPa (3-lb/in^2) positive pressure. The air volume in the storage tank shall be sufficiently large so that no breathing of the tank will occur as the result of a fluid temperature change between 16°C (60°F) and the normal operating temperature.

3.5.3.4 Storage tank material. The storage tank shall be fabricated from stainless steel and shall not be painted.

3.5.4 Fill and drain pump unit. An electric motor-driven pump shall be provided for each storage tank for filling and draining the steering system, steering system flushing, and emergency steering. The pump shall be in accordance with MIL-P-17869. The unit shall move one rudder at a rate of at least 1/2 degree per second, measured between 15 degrees right and left rudder angles, when the maximum actuator pressure is the MDWP. The unit shall be capable of continuously operating one rudder hard-over to hard-over when the maximum actuator pressure is 25-percent MDWP. The unit shall be capable of filling both head tanks of one rudder, from the low to the full marks, in 5 minutes or less. The unit shall be capable of flushing a rudder actuator, whether the actuator is being operated by the main pump or not.

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3.5.4.1 Control valves for emergency steering. A manually operated, manifold-mounted, directional control valve shall be provided for emergency steering operations, one for each rudder serviced by the fill and drain pump. Each valve shall control the rudder under aiding- and restoring-type rudder loads. The valve operating handle shall move easily and in the same direction as the rudder (a left movement of the handle for left rudder motion). The center position of the handle shall be detented, and the lines going to the rudder actuator shall be blocked and the pump bypassed in this position. Pilot-operated check valves shall be provided in each actuator line to prevent loss of rudder control during aiding-type rudder loads.

3.5.4.1.1 Other control valves. A manually operated, manifold-mounted, directional control valve shall be provided to direct the fill and drain pump flow to either rudder emergency control valve. An externally adjustable pump discharge relief valve shall be provided and shall be set to limit the fill and drain pump pressure to 110-percent MDWP.

3.5.4.1.2 Flushing function. Emergency control valve return pressure shall be appropriately elevated to provide the necessary replenishment pressure for the main pump when the fill and drain pump unit is flushing the rudder actuator while the actuator is being operated by the main pump. The flushing function shall not result in a reduction in the main pump head tank fluid level in this operating mode. The fill function may be used as appropriate to meet this requirement.

3.5.4.1.3 Portable hose connection. An auxiliary connection shall be provided in the pump discharge piping that shall include a quick disconnect fitting and suitably secured end cap for connecting a portable hose for flushing power unit piping. The hose shall be sized to accommodate the maximum fill and drain pump flow rate.

3.5.4.2 Suction strainer. The fill and drain pump unit shall take suction from the storage tank through a 50-mesh strainer. The strainer element shall be easily removable for cleaning and shall be arranged to minimize system contamination during removal.

3.5.4.2.1 Auxiliary suction connection. An auxiliary suction connection shall be provided and shall include a quick disconnect fitting and suitably secured end cap for connecting a portable hose for draining the supply tanks, the main pump cases, and the rudder cylinders. A valve shall be provided to isolate or partially isolate the storage tank from the pump suction when the auxiliary suction is used. Suitable hose stowage shall be provided.

3.5.4.3 Return line filter. A filter shall be provided to filter the return fluid from the emergency steering and flushing operations before it is returned to the storage tank.

3.5.4.3.1 Auxiliary filter discharge connection. An auxiliary connection shall be provided in the filter discharge piping and shall include a quick disconnect fitting, and suitably secured end cap for attaching a portable hose for filling the main pump cases and rudder cylinders. The auxiliary connection and piping shall permit filling of the rudder actuator cylinders from empty to full in less than 10 minutes.

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3.5.4.4 Minimizing heat generation. Fluid heating shall be minimized so that the heat dissipation of a 1/2-full storage tank, and the other components containing the heated fluid, will maintain the fluid temperature at a level no greater than 60°C (140°F) when filling, draining, or flushing continuously, or operating the rudder continuously at pressures up to 25-percent MDWP.

3.5.4.5 Fault light. A fault light shall be provided on the emergency steering unit motor controller remote start/stop control unit. The light shall activate when the pump pressure becomes excessive or when the storage tank temperature becomes higher than 75°C (167°F) ± 3°C (5°F).

3.5.4.6 Electric motor for fill and drain pump. The fill and drain pump electric motor shall be powered from the ship service emergency generator when the normal power sources fail. The motor shall not stall when all pump flow is passing over the pump discharge relief valves provided. The motor shall be in accordance with MIL-M-17060 and the requirements below:

Service: Navy A
Ambient temperature: 50°C (122°F)
Horsepower: as required to meet operating conditions without exceeding 150 percent of the nameplate rating
Design: as required
Enclosure: totally enclosed

3.5.4.6.1 Motor controller for fill and drain pump. The fill and drain pump motor controller shall be consistent with the fill and drain pump motor characteristics specified, and shall be provided in accordance with MIL-C-2212, and the following:

Duty - as required
Protective features - Overload and LVR protection shall be provided
Control voltage - 115 Vac, 60 Hz
Enclosure - drip-proof
Operation - Magnetic across the line

3.5.4.6.2 Control panel for fill and drain pump power unit. A remote panel shall be provided within clear view of, and easily accessible by, the emergency steering control valve operator. The panel shall comprise emergency steering pump unit motor START, STOP, and overload RESET switches. The panel shall also comprise MOTOR RUNNING, MOTOR STOPPED, and POWER AVAILABLE indicator lights, and the FAULT light specified herein. The switches and lights shall be identified on the panel using the capitalized nomenclature used herein. The switches shall be push button type.

3.5.4.6.2.1 RESET switch. Depressing the RESET switch shall reset the overload breaker after the overload heaters have sufficiently cooled. Sustained depression of the RESET switch shall override the overload breaker.

3.5.4.7 Resilient mounting. The fill and drain pump equipment, except the motor controller, shall be resiliently mounted using soundmounts in accordance with MIL-M-17308, and shall meet the airborne and structureborne noise requirements of 3.2.6.3 when the pump is operating in any of the modes specified in 3.5.4.

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3.5.5 Hand steering. A hand-operated pump shall be provided in parallel with the electric pump unit for tank filling and emergency steering operations when electric power is not available. The hand pump shall develop 3.1 MPa (450-lb/in^2) output pressure when the force on the pump operating handles is not greater than 90 N (20 pounds). The fluid volume pumped with each complete revolution of the crank handle shall be at least 41 ml (2.5 cubic inches).

3.5.5.1 Hand pump operation. The hand pump and the rudder directional control valve operating through the pilot-operated flow control valves shall operate and maintain control of the rudder under all steering gear operating conditions up to 3.1 MPa (450-lb/in^2) actuator pressure. The pump shall not be a safety hazard because of improper use or sudden external forces on the rudder.

3.5.6 Warning plates. Warning plates shall be provided at the emergency steering control valve and in the pilot house, stating that the ship speed and rudder angles must be limited so as not to exceed the specified emergency steering and hand steering pressures limitations when these equipments are used.

3.6 Controls. Functions of the motor controller such as sensor monitoring may be incorporated into the local control unit provided that all requirements specified herein are met.

3.7 Identification. Each item identified as a major component (see 3.2.1.1) shall have an identification nameplate affixed in a conspicuous location. Identification plates shall conform to MIL-P-15024, MIL-P-15024/5, MIL-STD-1472, and MIL-STD-130.

3.7.1 Identification nameplates. Identification of parts shall be as specified in MIL-STD-130, and the identification nameplate shall contain, as a minimum, the following information:

(a) Nomenclature.
(b) Part number.
(c) National stock number (if available).
(d) Serial number.
(e) Manufacturer's name and CAGEC.
(f) Contract number.
(g) Technical manual number (if available).
(h) Salient design characteristics (the pertinent rating, operating characteristics and other information necessary for identification of the item).

3.8 Technical data.

3.8.1 Drawings. Drawings shall be to scale and shall show details, dimensions, designating marks so that any part may be ordered therefrom (see 6.3). General arrangement drawings shall show sections, elevations, and plan views in detail to clearly indicate accessibility and clearances between parts.

3.8.2 Technical manuals. Equipment manuals shall be type I, in accordance with MIL-M-15071 (see 6.3). The manuals shall cover the complete steering machinery system with all modes of operation from all control stations.

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3.8.2.1 Operating section requirements. The operating section of the technical manuals shall include the following:

(a) The function of all control and indicating devices and the setup for each mode of operation.
(b) Proper sequence of operation of the controls, showing the setting required when transferring from one mode of operation to another.
(c) Operation and transfer of power units at all control stations.
(d) Interpretation of the indicating devices that may be observed during malfunction of the steering system and what must be done to correct the condition.

3.8.2.2 Maintenance section requirements. The maintenance section of the technical manuals shall explain the functions of the components in the system by use of a schematic diagram. The schematics and wiring diagrams furnished shall be suitable for use in maintenance of the system. Schematics of amplifiers and other electronic assemblies shall show all components within the assembly.

3.8.2.3 Drawings. A detail drawing shall be furnished in the technical manual for each component requiring adjustment and shall include the following:

(a) Complete operating parameters.
(b) Range of all adjustments.
(c) Settings required.
(d) Where, when, and how the adjustments are made.
(e) Voltage and current ranges (both ac or dc, where applicable) through which the equipment will successfully operate for all electrical components.
(f) Description of all internal and external connections; electrical, hydraulic, mechanical.
(g) Component drawings showing the component as installed in the system. Configurations not used within the system shall not be shown in the drawings.

3.9 System safety program. Safety design features, including fail-safe features, shall be incorporated into the design to prevent damage to equipment and to ensure optimal personnel protection during repair or interchanging of any component or assembly.

3.10 Workmanship. External finish shall be clean, smooth, and provide complete coverage. There shall be no signs of blistering, corrosion, discoloration, scrapes, gouges, or voids in the finish. Welds shall be free of harmful defects such as cracks, undercuts, voids, or gaps. There shall be no sharp or ragged edges, corners, burrs, or protrusions which might create a safety hazard to personnel. There shall be no dents or loose, damaged, bent, or broken parts or fasteners. There shall be defects that might reduce the reliable operational capability or life of the equipment. There shall be no signs of hydraulic fluid leakage. Piping and cabling shall be suitably supported by clamps or hangers. Cabling and wiring insulation shall not be nicked, gouged, cut, or otherwise damaged. Safety guards, covers, shields, and warning plates shall be appropriately installed and marked.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection and calibration system. The contractor shall prepare and maintain an inspection and calibration system.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Test conditions.

4.3.1 Test equipment accuracy. The accuracy of instruments and test equipment used to control the test parameters, whether located at a Government testing laboratory, the contractor's plant, or an independent testing facility shall be in accordance with MIL-STD-45662 for calibration cycle of specific instruments. Records of calibration history shall be maintained, and the latest calibration date shall be attached to each test instrument. Instruments and test equipment used in conducting the tests specified herein shall be as follows:

(a) Conform to laboratory standards whose calibration is traceable to prime standards at the U.S. National Bureau of Standards.
(b) Have an accuracy at least 1/3 the tolerance for the parameters to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any of the specific test methods, the specific test method requirement shall govern.

4.3.1.1 Sensors and their placement. Pressure transducers and recorders shall have the necessary frequency response to accurately reflect any pressure transients that may occur during the tests. The location of test transducers in the steering system components shall not degrade steering system performance.

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4.3.1.2 Monitored control voltages. Control voltages to be monitored shall be buffered to prevent interaction with the recording devices.

4.3.2 Temperature stabilization. Temperature stabilization shall be considered to have been attained when the temperature of any part of the test item considered to have the longest thermal lag does not change more than 1.7°C (3.0°F) per hour.

4.3.3 Test installation. The test item shall be installed in the test facility in a manner that will represent normal usage. Plugs, covers, and inspection plates not used in operation but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The test item shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test item in the facility is applicable only when operation is required during or immediately following the specific test.

4.4 First article inspection. First article inspection shall consist of examinations and tests, as specified in table I.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of examinations and tests, as specified in table I.

TABLE I. Examinations and tests.

Examination or test	Requirement	First article inspection	Quality conformance inspection
Visual examination	3.2.1 and 3.10	4.6.1	4.6.1
Surface examination	3.10	4.6.2	4.6.2
Workmanship	3.10	4.6.2	4.6.2
Design verification:			
Low temperature	3.2.1.4 and 3.2.1.4.1	4.7.2.1.1	-
High temperature	3.2.1.4 and 3.2.1.4.1	4.7.2.1.2	-
Vibration	3.2.6.2 and 3.2.6.2.1	4.7.2.2.1 and 4.7.2.2.2	-
Hydrostatic pressure	3.4.2.3 and 3.5.1.1	4.7.2.3	4.7.2.3
Shock	3.2.6.1	4.7.2.4	-
Air removal	3.5.2.6.1	4.7.2.5	
Main pump replenishment	3.5.2.1.4, 3.5.2.2 and 3.5.2.2.1	4.7.2.6	-
Tank inclination	3.5.2.7.5	4.7.2.7	-

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TABLE I. Examinations and tests - Continued.

Examination or test	Requirement	First article inspection	Quality conformance inspection
Steering system performance:			
Cooler	3.5.2.4	4.7.4.2.2	4.7.4.2.2
Auxiliary pump suction pressure	3.5.2.7.1	4.7.4.2.3	4.7.4.2.3
Power unit change-over	3.2.1.5, 3.2.1.2 and 3.5.2.3.2	4.7.4.2.4	4.7.4.2.4
Noise	3.2.6.3, 3.5.2.9 and 3.5.2.9.2	4.7.4.2.5	4.7.4.2.5
Storage tank	3.5.3.1, 3.5.3.3 and 3.5.4.4	4.7.4.2.6	4.7.4.2.6
Emergency steering	3.5.4.1, 3.5.4.6 and 3.5.5.1	4.7.4.2.7, 4.7.4.2.7.1, 4.7.4.2.7.2	4.7.4.2.7 4.7.4.2.7.1, 4.7.4.2.7.2
Actuator stops	3.4.4	4.7.4.2.8	-
Steady state voltage and frequency	3.2.1.4.2	4.7.4.2.9	-
Transient voltage	3.2.1.4.2	4.7.4.2.10	-
Transient frequency	3.2.1.4.2	4.7.4.2.11	-
Power interruption	3.2.1.5 and 3.5.2.3.2	4.7.4.2.12	-
Reliability	3.2.2.1	4.7.4.3 thru 4.7.4.3.7	4.7.4.3.7
Corrective maintenance	3.2.2.1	4.7.4.4	-
Preventative maintenance	3.2.2.1.3 and 3.2.2.1.3.1	4.7.4.5	-

4.6 Examinations.

4.6.1 Visual examination. When specified herein, the test item shall be visually examined and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, a visual examination of the test item within the test chamber shall be performed at test conditions, where possible. Upon completion of the test, a visual examination of the test item shall be performed again, and after the test item has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements shall provide reason to consider the test item as having failed to withstand the conditions of the test. Leakage or discoloration of impregnating compounds shall be considered damage and cause for rejection.

4.6.2 Surface examination. The machinery shall be externally examined to verify:

(a) Conformance to drawings with respect to the presence of all listed subassemblies, parts, and materials; drawing dimensions and tolerances; specified coatings and finishes.

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(b) That workmanship is as specified in 3.10. The equipment need not be disassembled for this examination except for the removal of covers and inspection plates.

The external finish shall be examined for cleanliness, smoothness, and coverage.

4.7 Tests.

4.7.1 Design verification tests. Prior to exceeding with any of the test methods, the test item shall be operated under the standard ambient conditions and a record made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test or both, as required.

4.7.1.1 Operation and performance checks. When operation of a test item is required during the test, the operation and performance checks shall be of sufficient duration, or shall be repeated at appropriate intervals, to ensure a record of comprehensive comparative data for comparison with data recorded under standard conditions.

4.7.2 Machinery subsystem tests. The following tests shall be accomplished, as a minimum, on the steering machinery prior to system integration.

4.7.2.1 Temperature tests.

4.7.2.1.1 Low temperature test. A low temperature test shall be conducted on one power unit. The power unit shall be locally controlled by a local control unit that is included in the test cell with the power unit. The test shall be in accordance with method 502.1 of MIL-STD-810, and as follows:

(a) Step 2 - Testing at the storage or nonoperating temperature is not required.
(b) Step 4 - Operating low temperature shall be 0°C (32°F).
(c) The combined duration of Steps 4 and 5 shall not be less than 12 hours, at least 2 hours of which shall be in the temperature-stabilized condition prior to conducting tests.

4.7.2.1.2 High temperature test. A high temperature tests shall be conducted on one power unit. The power unit shall be locally controlled by a local control unit included in the test cell with the power unit. The test shall be in accordance with method 501, procedure I of MIL-STD-810, and as follows:

(a) Step 4 - The high operating temperature shall be 49°C (120°F).
(b) The combined duration of Steps 4 and 5 shall not be less than 12 hours, at least 6 hours of which shall be in the temperature-stabilized condition prior to conducting tests.
(c) Step 7 - The equipment shall operate for a period of at least 2 hours in the temperature-stabilized condition before the Step 7 tests are performed. The power unit temperature regulating valve shall be adjusted to hold the tank temperature at nominally 52°C (125°F) at the stabilized condition. The performance test shall

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include a test with the cooling water temporarily shut off and the power unit fluid temperature allowed to rise. The test shall demonstrate that the power unit automatically shuts down and the CRITICAL FAULT light on the motor controller activates when the return fluid temperature reaches 85°C (185°F).

4.7.2.2 Vibration tests.

4.7.2.2.1 Power unit. A power unit shall be subject to the type I and type II vibration tests of MIL-STD-167-1 and shall be energized and operating during the test. The use of simulated loads, such as weights or other rigid dummy masses, is not acceptable. The test shall be performed with the normal assemblies in their functionally oriented positions. The power unit shall operate continuously and reliably during and after the vibration tests.

4.7.2.2.2 Control equipment. Control equipment, including the power unit motor controllers and the fill and drain pump motor controllers, shall be subjected to the type I vibration test of MIL-STD-167-1. The equipment shall be operated during the test. Out-of-tolerance operation of the equipment during or after the test, which results in the steering gear not meeting its performance requirements, will require redesign of the faulty part and retesting until all test conditions are met.

4.7.2.3 Hydrostatic pressure test. A power unit shall be subjected to a hydrostatic pressure test. The components which may be subjected to the MDWP during operation of the main hydraulic pump shall be hydrostatically tested to 34.5 MPa (500 lb/in^2). Other components shall be hydrostatically tested to 1.5 times their maximum working pressure. No damage or leakage shall be evident during or as a result of these tests, and the power unit shall operate within specified performance limits after the tests. Sheet plastic or other suitable material shall be placed over the supply tank suction strainer to facilitate developing the pressure differential across the screen. Test fluid shall be completely removed from the system after the tests if it is not the normal system fluid.

4.7.2.4 Shock test. A power unit shall be subjected to the grade A, class II shock test of MIL-S-901. The fill and drain pump equipment shall be subjected to a grade A, class I shock test. The use of simulated loads, such as weights or other rigid dummy masses, is not acceptable. The test shall be performed with the equipment mounted in its functionally operating position, but the power unit need not be operating during the test. Equipment shall operate within specified performance limits after completion of the shock test.

4.7.2.5 Air removal test. A power unit shall be tested to demonstrate that the supply tank removes at least 90 percent of a volume of air bled into the tank inlet fluid. Air shall be injected through a no greater than 0.3 mm (0.012-inch) inside diameter syringe needle installed perpendicular to and extending to near center of the flow in the tank inlet piping. Soft plastic tubing having the approximate size of the tank inlet piping may be substituted for portions of the piping to permit insertion of the needle for this purpose. An air volume equal to at least 5 percent of the tank oil volume shall be bled into the tank inlet fluid at not greater than 1/4 percent of the tank inlet flow rate, by volume. The

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specified air volumes shall be at fluid pressure. The supply tank vent piping may be disconnected from the head tank for measuring the amount of scavenged air. This test shall be conducted with the fluid within the normal operating temperature range and after the power unit has been sufficiently long to ensure that all entrained air initially in the fluid has been removed.

4.7.2.6 Main pump replenishment test. A power unit shall be tested with the means provided to draw flow out of one of the main pump lines at no less than 2/3 the maximum flow rate of the pump. The pump shall be running with the swashplate held at zero stroke. The test shall demonstrate that the replenishment accumulator piping and replenishment check valves are adequately sized, so that fluid in the main pump lines does not fall below the pressure, measured at the pump output flange, that the pump manufacturer certifies is sufficient to prevent main pump cavitation. This test shall be conducted with the fluid at the normal operating temperature and with only one replenishment check valve active.

4.7.2.7 Tank inclination test. A head tank shall be tested to demonstrate that when the fluid level in the tank is at the low level mark on the tank level gage, the empty tank level switch will not be activated when the head tank is inclined, as specified in 3.2.1.4.

4.7.3 Control subsystem tests. Testing of the steering control subsystem equipment shall be in accordance with requirements in the ship specification and as specified herein.

4.7.4 Steering system general performance tests.

4.7.4.1 Test setup. Two power units, including the motor controllers and their dedicated local control units, a storage tank, and a fill and drain pump, shall be arranged relative to each other in the test facility as they will be arranged onboard ship. The Bridge Steering Unit and Aft Steering Unit shall be located in the area for steering system control. A rudder load simulator shall be provided comprising two hydraulic actuators, gas precharged accumulators, and necessary valves. The rudder load simulator shall be connected to the power units using the same piping and configuration to be supplied onboard ship.

4.7.4.1.1 Load-simulating actuators. The hydraulic actuator connected to the power units (power unit actuator) shall have a displacement volume equal to the displacement volume of the ship rudder actuator assembly. The power unit actuator piston rod shall be mechanically attached to the piston of the second actuator (load actuator) whose ports are individually connected to gas precharged accumulators.

4.7.4.1.2 Load-simulating accumulators. The pressure developed by the test unit shall be controlled by the accumulator gas and oil precharge and the movement of the load actuator piston. The rudder load simulator shall subject the active power unit to aiding and restoring load pressures from nominally zero up to and including the MDWP while the pump is operating at all displacements between zero and maximum.

4.7.4.1.3 Rudder angle transmitter unit drive linkage. The rudder angle transmitter unit drive linkage, designed for use with the load simulator, shall operate the test unit rudder angle linkages as they will operate on the ship.

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4.7.4.2 Test procedures. Test procedures shall be prepared to demonstrate that the requirements for the following, as a minimum, have been met:

(a) Average rudder rate, positional accuracy, control stability.
(b) MDWP and main pump piping/manifold loss.
(c) Remote and local power plant starting/stopping, interlocking, control station transfer, and steering mode selection.
(d) Transfer valve and pump motor performance under aiding and restoring loads with electric power loss.
(e) Pump stroke time and control stability.
(f) Fault indicator and alarm activation for system faults.
(g) Machinery noise levels.
(h) Fill and drain/emergency steering/flushing pump unit performance.
(i) Hand pump performance.

These test procedures shall be implemented using the steering system rudder load simulator. Pressure and temperature transducers and selected control equipment voltages shall be connected to continuous recorders to demonstrate and document steering system performance. Failure to perform properly in any condition, including inadvertent activating of fault displays, creating of potential safety hazards, or evidence of potential steering system damage shall each constitute failure to meet performance requirements.

4.7.4.2.1 Test facility temperature. Unless otherwise specified herein, the ambient temperature of the test facility space and the initial steering equipment temperatures shall be between 16 and 49°C (60 and 120°F). When the normal operating temperature condition is specified the system shall operate with the supply tank temperature at 52°C, ± 3°C (125°F, ± 5°F).

4.7.4.2.2 Cooler performance. Tests shall be conducted to demonstrate that the cooler meets the requirements specified in 3.5.2.4 with the cooling water inlet temperature as high as 32°C (90°F).

4.7.4.2.3 Auxiliary pump suction pressure. During power unit startup, tests shall demonstrate that the suction pressure may become transiently negative, but not sufficient to cause auxiliary pump audible cavitation when the head tank is at the low level mark.

4.7.4.2.4 Power unit changeover. A test shall be conducted to demonstrate that, with the rudder load simulator moving and not moving and with loads up to and including maximum load, the standby power unit can be started and brought on-line within 4 seconds, from the Bridge Steering Unit and Aft Steering Unit. The test shall demonstrate that the LVR and LVP transfer requirements for active and standby pumps as specified in 3.5.2.3.2 are met, and that the standby pump can be operated from the motor controller, with or without the other pump in active status. The test shall also demonstrate that when the head tank is intentionally emptied, the standby pump is automatically started and assumes active status.

4.7.4.2.5 Noise. Structureborne noise levels shall be measured in accordance with MIL-STD-740-2 while the rudder load simulator is being cycled and operating up to MDWP. One-third octave noise levels shall be successively recorded at not greater than 1/4-second intervals over at least a 1-second period that shall bracket the noisiest combination of ram position and pump stroke

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obtained using maximum helm angle step commands. The noisiest level shall be at or below the specified maximum noise levels. Performance shall be documented after the system fluid temperature is at the normal operating temperature.

4.7.4.2.6 Storage tank. Tests shall be conducted to demonstrate that:

(a) Maximum design storage tank pressure is not exceeded under all operating conditions.
(b) The storage tank fluid temperature is maintained at a temperature not greater than 60°C (140°F) when the fill and drain pump is run continuously for up to 8 hours in the flushing mode without the power unit operating.
(c) The storage tank air relief valve assembly prevents tank breathing when the tank temperature changes between ambient and normal operating levels.
(d) The fill and drain pump does not cavitate when the fluid level in the storage tank is at the low low level.

4.7.4.2.7 Emergency steering. A test shall be conducted with the rudder load simulator adjusted for 25-percent MDWP at full displacement. The test shall demonstrate that:

(a) The rudder can be satisfactorily controlled and that the actuators operate smoothly under all loads and operating directions.
(b) The fluid temperature is maintained at a temperature not greater than 60°C (140°F). This test shall be conducted by continuously cycling the emergency steering control valve to operate the rudder load simulator hard-over to hard-over until the fluid temperature stabilizes.

4.7.4.2.7.1 Actuator flushing test. The rudder actuator shall be flushed with the fill and drain pump while the main pump is operating the rudder. The test shall demonstrate that this can be accomplished at all rudder loads and rates up to 2/3 maximum ship speed, and that the main pump head tank fluid level stays constant in this operating mode. The test shall also demonstrate that the fill and drain pump motor does not stall at 110-percent MDWP.

4.7.4.2.7.2 Hand pump test. The rudder load simulator shall be operated by the hand pump, with the rudder load simulator adjusted to produce 3.1 MPa (450-lb/in^2) maximum pressure at the hand pump discharge port. The test shall demonstrate that:

(a) The hand pump can be brought on-line and that the rudder actuators can be satisfactorily controlled under all aiding and restoring loads up to 3.1 MPa (450-lb/in^2) hand pump pressure.
(b) No safety hazard exists when the pump is operated with the actuator pressure at the MDWP.

4.7.4.2.8 Ram overtravel stops. The contractor shall configure the rudder load simulator to include the crosshead copper stops and steel hardstops in the assembly. The stops shall be supported as rigidly as they will be on the ship, and the rudder load simulator assembly shall be appropriately restrained, while conducting this test. Tests shall be conducted to demonstrate that the copper

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stops will remain effective after being impacted not less than twice. The pump shall be at full displacement and the rudder load simulator shall be at a displacement and moving in a direction that produces an aiding load equal to at least 50-percent MDWP at the time of impact. This test shall demonstrate that there are no excessive pressure transients in the main pump output lines and pumpcase, rudder actuator, and the supply tank. The test shall also demonstrate that the pump is immediately returned to nominally zero displacement but maintains pressure at nominally 110-percent MDWP, and that the main pump motor does not stall after the impact has occurred. The same results shall be obtained after the test is successively repeated a third time, but with the rudder load simulator producing a restoring load equal to the MDWP at the time of impact. Failure of the copper stops during the third impact test is allowed, but the steel hardstops shall remain functional.

4.7.4.2.9 Steady state voltage and frequency. The steering system shall be operated for at least 15 minutes in each of the following conditions:

(a) Reference condition - normal voltage, normal frequency.
(b) 90-percent normal voltage, 95-percent normal frequency.
(c) 110-percent normal voltage, 95-percent normal frequency.
(d) 110-percent normal voltage, 105-percent normal frequency.
(e) 90-percent normal voltage, 105-percent normal frequency.

4.7.4.2.10 Transient voltage. The steering system shall perform normally after being subjected to the following conditions:

(a) With the control system operating at a steady state voltage 110 percent of normal, a transient voltage of 118 percent of normal recovering to the steady state voltage within 2 seconds shall be superimposed.
(b) With the control system operating at a steady state voltage 90 percent of normal, a transient voltage of 82 percent of normal recovering to the steady state voltage within 2 seconds shall be superimposed.

4.7.4.2.11 Transient frequency. The steering system shall perform normally after being subjected to the following conditions:

(a) With the control system operating at a steady state frequency 105 percent of normal, a transient frequency of 108 percent of normal recovering to the steady state frequency within 2 seconds shall be superimposed.
(b) With the control system operating at a steady state frequency 95 percent of normal, a transient frequency of 92 percent of normal recovering to the steady state frequency within 2 seconds shall be superimposed.

4.7.4.2.12 Power interruption. With the steering system operating within the steady state tolerances of voltage and frequency, the external power supply shall be suddenly interrupted. After an interval between 3 and 4 seconds, the power supply, within the steady state tolerances, shall be reapplied. After the system has been operated long enough to detect any major performance degradation, the power shall be interrupted for an interval of 29 to 30 seconds. This

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procedure shall be conducted at least 5 times. As a result of this test, no damage to the steering system shall be incurred, and the steering system shall operate normally after the test. The power interruption test shall be repeated for all system modes of operation.

4.7.4.3 Reliability test. The power units for one rudder shall be subjected to a reliability development growth test. One power unit shall be selected as the active power unit and shall be operated for at least 1000 hours total during the reliability test. The test shall be in accordance with MIL-STD-785, Task 302 and as specified in 4.7.4.3.1 through 4.7.4.3.6.

4.7.4.3.1 Simulators. During the reliability test, the rudder load simulator specified in 4.7.4.1 shall be set up to produce the MDWP at the maximum stroke of the rudder load simulator, and the accumulator gas volume shall be selected so that the rudder load simulator produces at least 40-percent MDWP when the stroke is 50-percent maximum. The power units shall be controlled to simulate sudden hard-over to hard-over helm orders at the following frequency:

$$F = 1/(2 + 4MA/MR)$$

Where: F - frequency (Hz)
MA - maximum specified rudder angle (degrees)
MR - minimum specified rudder rate (degrees/per second)

4.7.4.3.2 Test conditions. The fluid temperature during the reliability test shall be at normal operating temperature and the test facility temperature shall be between 16 and 49°C (60 and 120°F). The active pump shall be stopped and started at least four times each day. The standby pump shall be started and switched to be the active unit for 30 seconds at least once a day. Equipment doors and covers that are normally closed in service shall be closed for this test. Operating temperature within control components shall be sampled during the test.

4.7.4.3.3 Preventive maintenance. Only that preventive maintenance proposed by the contractor for the technical manuals and approved by the Government prior to the test shall be performed during this test.

4.7.4.3.4 Pump teardown inspection. At the completion of the reliability test, a power unit pump shall be disassembled and inspected for signs of impending failure or excessive wear.

4.7.4.3.5 Mean-time-between-failures. Failures that affect the steering system equipment MTBF ratings during the reliability test shall be defined as follows:

(a) Failures that require a shutdown of the steering system due to equipment failure or overheating, or create a hazard to personnel or equipment safety.
(b) Failure of the steering system to respond to signals from the Bridge Steering Unit, Aft Steering Unit, or local control unit.
(c) Degradation of steering system operation below the performance levels specified in 3.2.1.1.
(d) Any pattern failure (for example, a repetitive failure) of the same or similar components performing the same or dissimilar functions.

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(e) An impending failure which, in the opinion of the Government, would result in a failure if operation were continued for a year without corrective maintenance.
(f) If post teardown reveals signs of wear that are not consistent, in the opinion of the Government, with the MTBF requirements as specified in 3.2.2.1.
(g) Any need for PM not previously approved by the Government.
(h) External leakage of hydraulic fluid in excess of 1 milliliter per hour.
(i) The need for or the performance of a corrective maintenance action requiring in excess of 3 hours to accomplish.

4.7.4.3.6 Reporting of failures. Failures occurring during the reliability test shall be reported, analyzed, and corrected. Corrective actions shall be proposed to and approved by the Government, and shall be reflected in the drawings and technical manuals after approval.

4.7.4.3.7 Failure-free operating test (burn-in). In addition to the tests specified in 4.6.2, 4.7.2.2 and 4.7.2.3, each production steering system shall be subjected to a 100-hour burn-in test in the reliability test mode, the last 48 hours of which shall be failure free. Testing shall be in accordance with procedures for Environmental Stress Screening set forth in MIL-STD-785, Task 301. If a failure occurs during the last 48 hours of the test, the test shall continue until the equipment has logged a minimum of 48 consecutive failure-free hours of operation. Upon successful completion of 48 consecutive failure-free hours of operation, the equipment shall be subjected to all of the performance test requirements. Failure to meet any of the performance test requirements shall require repeat of the failure-free portion of the burn-in test until all test conditions are met.

4.7.4.4 Corrective maintenance test. Thirty corrective maintenance tasks shall be performed on the steering system to demonstrate that the mean-time-to-repair requirements are met. The Government will participate in the task selection. The tasks shall include replacing main and auxiliary pumps, replacing the main pump drive shaft seal, replacing the pump stroke feedback potentiometer assembly, and troubleshooting selected control subsystem failures. The controls tasks and testing shall be in accordance with MIL-STD-471. Only the corrective maintenance procedures described in the technical manuals shall be used. Fault localizing, fault isolating, part removal, part repair, part replacement, and verification of performance times shall be recorded.

4.7.4.5 Preventative maintenance test. At least 10 PM tasks shall be performed on the steering gear machinery. The Government will participate in the task selection. Replacement of all filter elements, without excess spillage and without the need to drain the tanks, shall be demonstrated. Removing of the storage tank strainer without introducing contamination downstream of the strainer shall be demonstrated. The time to perform these tasks shall be weighted according to the frequency noted in the technical manuals. The weighted PM shall meet the requirements specified in 3.2.2.1.3.

4.8 Test plan. The contractor shall furnish a written examination and test plan (see 6.3). The test plan shall contain the procedures, data forms, a list of test equipment to be used, the sequence of tests, and the planned dates for performing inspections.

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4.8.1 Test procedures. Test procedures shall contain step-by-step information for each inspection, test, and demonstration to be performed. The test procedures shall include preparation for test connections, disconnections, adjustments, and measurements.

4.8.2 Test equipment list. The test equipment list shall identify the test equipment to be used by name, model, range, and accuracy. Special test equipment shall also be listed and identified to describe its function in performance of the test.

4.8.3 Test data forms. Test data forms shall be prepared for each of the procedures. The forms shall be organized to indicate the acceptance limits of the test and the actual readings observed. In addition, the forms shall indicate a statement of acceptance or rejection after each test. The forms shall contain a signature and date block for certification of the data by the test engineer.

4.9 Inspection reports. Inspection reports containing results of tests shall be prepared (see 6.3). When the tests are performed, the test data and test data forms shall comply with requirements as specified herein. Any failure occurring during the tests or as a result of the tests specified herein shall be included in the report together with the probable cause of failure and corrective action needed to preclude recurrence.

4.10 Inspection records. Inspection records of the examinations and tests shall be kept complete and available to the Government.

4.11 Inspection of packaging. Sample packs, and the inspection of the preservation, packing, and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Preservation, packing, and marking. The electrohydraulic steering system shall be preserved, packed, and marked in accordance with ASTM 3951 and as follows:

(a) The hydraulic equipment shall be cleaned and protected (openings sealed or capped) prior to shipment in accordance with MIL-STD-419.
(b) Drawings shall be folded not to exceed 8.5 x 15 inches.
(c) Microfilm aperture and tabulating cards shall be packed within a close fitting regular slotted-type fiberboard box.
(d) Technical manuals which accompany shipments shall be packaged in a transparent waterproof plastic bag, minimum four mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed with any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall state the

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approximate location therein. The manual shall be readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specifications.

6. NOTES

6.1 Intended use. This specification is intended to be applicable to any electrohydraulic steering system where such systems combine an electric motor driving one or more hydraulic pumps which provide hydraulic fluid under pressure to a hydraulic actuator that in turn is mechanically linked to the ship rudder or rudders.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type of steering system required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.1).
(e) Maximum rudder angle (see 3.2.1).
(f) Rudder rate (see 3.2.1).
(g) Maximum rudder torque, restoring and aiding, ahead and astern (see 3.2.1).
(h) Reliability program plan, if required (see 3.2.2.1).
(i) Maintainability program plan, if required (see 3.2.2.1).
(j) Maximum size and weight of steering system (see 3.3.2).
(k) Number of rams per rudder (see 3.4.1).
(l) When the roll stabilization option is required (see 3.5.2.3.2.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.2.1	DI-R-7079	Reliability program plan	MIL-STD-785
3.2.2.1	DI-MNTY-80822	Maintainability program plan	---
3.2.2.1.1	DI-L-7114	Logistic Support Analysis Strategy Report	MIL-STD-1388/1

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.2.1.3	DI-MNTY-80994	Planned maintenance system (PMS) functional block diagram	---
3.2.9	DI-MISC-80296	Design data and calculations	---
3.2.10	DI-L-2101	Standard, technical and maintenance overhaul and repair	---
3.2.11	DI-ILSS-80521	Material Status Report	---
3.2.12	DI-ILSS-80957	Integrated allowance document (IAD)	---
3.8.1	DI-DRPR-81000	Product drawings, engineering and associated lists, as follows: All steering machinery system production drawings Steering Machinery System Installation Drawings Steering Machinery System Electrical Schematic Steering System Hydraulic Schematic Steering Machinery System Wiring Diagram Steering System Piping Diagram Steering System Motor Controller and Transfer Valve Wiring Diagram	---
3.8.2	(See 6.4.1)	Manual, technical, manuscript copy as follows:	MIL-M-15071
	(See 6.4.1)	Machinery subsystem manual	Type I
3.9	DI-SAFT-80102	Safety assessment	---
4.8	DI-T-23731	Notification of tests	---
4.8	DI-NDTI-80603	Test Procedure	---
4.9	DI-T-2072	Reports, test	---
4.9	DI-MNTY-80824	Data collection, analysis and corrective action system, reports	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements

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Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Technical manual contract requirement. Technical Manual Contract Requirement (TMCR) is an attachment to a contract specifying technical manual development requirements.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Definitions.

6.6.1 Aft Steering Unit (ASU). Console located in the steering gear room and the equipment contained therein. The console is arranged similarly to steering panels on the bridge and includes (but is not limited to) the steering wheel, angle indicators, switches, status indicators and other equipment needed to sustain complete control of the steering system under all operating modes.

6.6.2 Bridge Steering Unit (BSU). Steering control equipment located in the ship control console on the bridge including (but not limited to) a steering wheel, angle indicators, alarms, and equipment needed to interface with other units in the steering control system.

6.6.3 Local control unit (LCU). Unit in the steering control subsystem which provides the control signals to the electrohydraulic servo valve on the pump. LCUs provide power to the rudder angle transmitter units and receive electrical rudder position feedback for processing with the command signals. LCUs and motor controllers are the primary interfaces between the control and machinery subsystems. LCUs also provide emergency steering capability through potentiometers located on their front panels.

6.6.4 Rudder angle transmitter unit (RATU). Unit containing dual potentiometers and a synchro which supplies the control subsystem with electrical signals proportional to the actual rudder position.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-17803E(SH)	

3. DOCUMENT TITLE
STEERING SYSTEM, ELECTROHYDRAULIC, MARINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
Technical Point of Contact (TPOC): Mr. W. Wilcox (SEA 56W3) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 703-602-1595
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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